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Original article

Plasma exosomal RNAs have potential as both clinical biomarkers and therapeutic targets of dermatomyositis

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Abstract

Objectives. DM is characterized by skeletal muscle weakness and cutaneous manifestations. Plasma exosomes (EXOs) contain proteins, RNAs, DNA, and lipid cargoes and are transferred among cells. If thoroughly investigated, plasma EXO RNAs could potentially improve our understanding of DM pathogenesis. We aimed to identify potential new biomarkers and therapeutic targets for DM.

Methods. The RNA (mRNA, miRNA and lncRNA) profiles of plasma EXOs were evaluated by sequencing on the Illumina HiSeq 3000 platform. Differentially expressed (DE) RNAs and bioinformatic analyses were performed. Human skeletal muscle myoblasts cells (HSkMCs) were stimulated with plasma EXOs, rapamycin or IFN-β. Real-time PCR and western blot analysis were used to detect related genes and proteins.

Results. A total of 689 DE mRNAs, 53 DE miRNAs and 452 DE lncRNAs were identified in DM plasma EXOs. Bioinformatic analysis inferred that plasma EXOs were secreted mainly by CD8+ T cells, regulatory T cells and natural killer cells. The DE miRNAs participated in the autophagy, TGF-β and Wnt signalling pathways. Three DE miRNAs (hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p) were correlated with serological indices, organ involvement and myositis-specific autoantibodies. The DE lncRNAs participated in autophagy, IFN-β production and mTOR signalling. DM plasma EXOs can induce autophagy in HSkMCs by regulating three miRNAs (hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p) and three lncRNAs (ENST00000584157.1, ENST00000523380.1 and ENST00000560054.1), which formed an autophagy network, playing a role in muscle damage.

Conclusion. Our study provides an overview of distinct RNA profiles in DM plasma EXOs, and verified some miRNAs as potential biomarkers and therapeutic targets. The findings provide important clues for more in-depth explorations of plasma EXOs in DM.

Key words: dermatomyositis, plasma exosome, mRNA, miRNA, lncRNA, sequencing, bioinformatics analysis, autophagy, biomarkers

BASIC SCIENCE

Rheumatology key messages

- Many differentially expressed mRNAs, miRNAs and lncRNAs were identified in dermatomyositis plasma exosome profiles.
- Three differentially expressed miRNAs were correlated with serological indices, organ involvement and myositis-specific autoantibodies.
- Dermatomyositis plasma exosomes can induce autophagy in HSkMCs by regulating 3 miRNAs and 3 lncRNAs.

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Introduction

DM is a subtype of idiopathic inflammatory myopathy that is characterized by chronic inflammation of skeletal muscle, leading to muscle weakness. Other organs, including skin (heliotrope rash, periorbital oedema and Gottron's papules), joints, heart, lungs and gastrointestinal tract are frequently affected, contributing to morbidity and mortality [1].

The mechanisms of skeletal muscle damage in DM remain largely unknown. Genetic factors, environmental factors (ultra-violet radiation, smoking) and infections have been implicated in the aetiology of DM. Immune mechanisms (such as lymphocyte-mediated cytotoxicity, autoantibody activity, overexpression of MHC-I on the muscle sarcolemma, and cytokine and chemokine signalling) have long been recognized as contributing to muscle weakness and damage [2–4]. In recent studies, the importance of non-immune mechanisms such as epigenetics has become increasingly apparent in the pathogenesis of DM. Certain miRNAs and long non-coding RNAs (lncRNAs) have been identified as biological markers for the diagnosis and prognosis of DM.

Exosomes (EXOs) are spherical membrane vesicles released by all cells; they are 30–150 nm in diameter, and express surface markers, including tumour susceptibility gene 101, integrins and tetraspanins (CD63, CD9 and CD81/82) [5]. They can carry proteins, lipids and nucleic acids as cargoes and participate in various physiological and pathological processes. RNAs (mRNAs, miRNAs and lncRNAs) in serum EXOs can be delivered into distant cells, and participate in cell communication, antigen presentation, and immune regulation [6]. Serum EXOs can restore cellular function *in vitro* and can be employed for the treatment and non-invasive diagnosis of dysferlinopathy [7]. EXOs released from inflammatory C2C12 myotubes can regulate myocyte differentiation and myokine expression *in vitro* [8].

In JDM patients, endothelial cells can take up plasma EXOs and alter their transcriptome, suggesting a previously unsuspected mechanism underlying the targeting of the vasculature in JDM [9]. However, the roles of plasma EXOs have not been reported. Here, we first investigated the profiles of RNAs (mRNA, miRNA and lncRNA) in DM plasma EXOs, then analysed the correlations of some altered EXOs-RNAs with clinical indices. Finally, we explored the roles of EXOs-ncRNAs in skeletal muscle damage, aiming to identify potential new biomarkers and therapeutic targets of DM.

Materials and methods

Patients and controls

Blood samples were collected from 22 normal controls (NCs) and 32 DM patients (including 23 before treatment and 9 after treatment). The DM patients before treatment group included patients who had just been diagnosed and who had not commenced any treatment. The DM patients after treatment group included patients who had been definitively diagnosed and treated. After the first sample collection, the DM patients started regular antirheumatic therapy (with prednisone, CYC, MTX, thalidomide, etc.). All participants were Han Chinese. All DM patients were enrolled in Xiangya Hospital of Central South University and fulfilled the 2017 EULAR/ACR classification criteria [10]. The clinical characteristics of all participants, including the demographic characteristics, serological characteristics, organ involvement and medications used, are shown in Supplementary Table S1, available at *Rheumatology* online. From among the participants, we carefully chose five newly diagnosed DM patients before treatment and five age- and sex-matched NCs for plasma EXO mRNA, miRNA and lncRNA sequencing.

The study was approved by the Ethical Committee of Xiangya Hospital of Central South University (201404360) in accordance with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. All blood samples were obtained with the written informed consent of patients and NCs, and the data were processed anonymously.

Isolation and identification of plasma EXOs

Peripheral whole blood samples (10 ml) were collected in EDTA anticoagulant-coated tubes and processed within 2 h. We isolated plasma EXOs using an exoEasy Maxi Kit (Qiagen, Frederick, USA) according to the manufacturer's instructions. Finally, we obtained 600 μl of EXO buffer from 4 ml of plasma per sample. We identified plasma EXOs by dynamic light scattering with a Zetasizer Nano ZS instrument (Malvern Instruments, Marvin, UK), and by transmission electron microscopy (TEM) (FEI Tecnai G2 Spirit Twin, Hillsboro, USA) with a negative staining technique. We also verified the expression of the EXO biomarker CD63 with a FACSCalibur flow cytometer (BD Biosciences, San Jose, USA) using EXOome-Human CD63 Isolation/Detection Dynabeads (Invitrogen, Thermo Fisher Scientific, Lithuania) and anti-human CD63-PE antibodies (12–0639, eBioscience, USA) according to the manufacturer's protocol.

Library construction, sequencing and data processing of plasma EXO RNAs

We isolated total RNA from plasma EXOs with TRIzol (Invitrogen Life Technologies, California, USA) according to the manufacturer's protocol. We constructed a miRNA library with an NEBNext® Multiplex Small RNA Library Prep Set for Illumina (NEB, Ipswich, USA) and constructed the mRNA and lncRNA libraries with an NEBNext® Ultra™ RNA Library Prep Kit for Illumina (NEB) according to the manufacturer's instructions.

Paired-end sequencing (PE150) on the Illumina HiSeq 3000 platform and data processing were performed at Guangzhou RiboBio Co. (Guangzhou, China). In brief, FastQC software and Trimmomatic tools (v 0.36) were used to obtain clean reads from sequencing reads via

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preprocessing. The mapped reads were obtained by mapping the clean reads to the human reference genome hg19; Burrows-Wheeler Aligner (BWA, v0.7.12) was used for small RNAs, and HISAT2 (v2.1.0) and HTSeq (v0.6.0) were used for mRNAs and lncRNAs. The miRDeep2 database was used to identify known mature miRNAs based on miRBase (v21) (http://www.miRBase.org) and to predict novel miRNAs.

Bioinformatic analysis for differentially expressed mRNAs, miRNAs and lncRNAs

To identify the significantly differentially expressed (DE) genes, the R package edgeR was used for mRNAs (with threshold $P < 0.05$ and fold change > 1.5) and miRNAs (with threshold $P < 0.05$ and fold change > 2.0), while R package DESeq2 was used for lncRNAs [with threshold $P < 0.05$ and absolute value [$\log_2$(fold change)] > 1.5). The R package ggplot2 was used for volcano plot construction and hierarchical clustering analyses.

To explore which kinds of immune cells the plasma EXOs come from, we used CIBERSORTx to enumerate the original immune cell proportions. CIBERSORTx is a previously published machine-learning method for calculation of the abundance of individual cell type bulk transcriptomes [11]. This method has been extensively used in calculating the immune cell proportions in tumour tissues. Here, we used LM22 [12] (a microarray-derived signature matrix for distinguishing 22 human haematopoietic cell subsets) and deconvoluted the mRNA profiles of the plasma EXOs.

To predict the target genes of selected miRNAs, we used the TargetScan, miRDB, miRTarBase and miRWalk software programs and selected only target genes that were predicted by at least three databases. For prediction of target genes of selected lncRNAs, target genes in cis were identified by integrating the sets of DE lncRNAs and their adjacent (within 10 kb) DE mRNAs, and target genes in trans were identified by both BLAST software (for initial screening, $e < 1E\text{-}5$) and RNAplex software (for re-screening). Moreover, we used the miRanda, PITA and RNAhybrid software programs to identify elements common to selected miRNAs and lncRNAs, and to predict miRNAs associated with lncRNAs of interest.

The DE mRNAs and target genes of the DE miRNAs and DE lncRNAs were subjected to pathway analysis with the Gene Ontology (GO) (http://geneontology.org/) and Kyoto Encyclopedia of Genes and Genomes (KEGG) (https://www.genome.jp/kegg) databases. GO and KEGG enrichment analyses were performed with the R package clusterProfiler [13]. Interaction networks for lncRNAs, miRNAs and mRNAs were constructed with Cytoscape software (http://apps.cytoscape.org/).

miRNA and lncRNA microarray data processing

Raw data from the GSE94535 dataset were downloaded from the Gene Expression Omnibus (http://www.ncbi.nlm.nih.gov/gds/). The R package limma was used for statistical testing. DE miRNAs were defined as those with an adjusted P-value of <0.05 and a fold change of >2.0. The lncRNA expression data in DM muscle tissue were provided by the China–Japan Friendship Hospital [14].

Cell culture and treatment

Human skeletal muscle myoblasts cells (HSkMCs) were purchased from ScienCell and cultured in skeletal muscle cell medium (ScienCell, San Diego, USA) supplemented with 5% EXO-free fetal bovine serum (FBS, Gibco), which was prepared by collecting the supernatant after ultracentrifugation of FBS at 150 000 $\times g$ for 3 h at 4°C (the same procedure was used for the EXO-free FBS described below). HSkMCs were cultured at 37°C in a 5% CO_2 humidified incubator and split at 60–65% confluency before the myoblasts contacted one another. We stimulated HSkMCs with 100 ng/ml IFN-β (300-02BC-5, PeproTech, USA) [15] or 60 nM rapamycin (V900930, Sigma-Aldrich, USA) [16] and control agents for 24 h and cultured them with 20% plasma EXOs (volume concentration, e.g. 160 μl of neutrophil EXO buffer and 640 μl of medium per well in six-well plates) from DM patients and NCs for 24 h. Cells were harvested to obtain RNA and protein. All experiments were repeated three times.

Real-time quantitative PCR

Total RNA extracted from plasma was analysed in an miRNEASY Serum/Plasma Advanced Kit (Qiagen, Frederick, USA) with a spike-in control RNA (cel-miR-39-3p standard RNA, RiboBio, Guangzhou, China) according to the manufacturer's instructions. Total RNA was extracted from HSkMCs using TRIzol according to the manufacturer's instructions.

The relative expression levels of miRNAs were measured with a miDETECT A Track™ miRNA qRT-PCR Starter Kit (RiboBio, Guangzhou, China) in a 7500 Real-Time PCR System. We purchased miRNA-specific forward primers and the Uni-Reverse Primer from RiboBio. The cel-miR-39-3p was used as the exogenous control for plasma, and U6B small nuclear RNA (*RNU6B*) was used as the endogenous control for HSkMCs for normalization of sample data.

For lncRNAs and mRNAs, we used a PrimeScript™ RT Reagent Kit (Perfect Real Time, Takara, Japan) to prepare cDNA and used TB Green PCR Master Mix (Takara, Japan) to perform qPCR in the 7500 Real-Time PCR System (Applied Biosystems, California, USA). *GAPDH* expression was used as the endogenous control for normalization of sample data. The gene-specific primer information is provided in Supplementary Table S2, available at *Rheumatology* online.

Western blot analysis

For protein analysis, 20 μg aliquots of total protein were separated by 8% or 15% SDS–PAGE and transferred to polyvinylidene fluoride (PVDF) membranes (Immobilon-P^{SQ}

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Transfer Membrane, Millipore, USA). Then, membranes were blocked with 5% skim milk powder in Tris-buffered saline containing 0.1% Tween (TBST) for 2 h at room temperature and incubated with anti-GAPDH (1:1000, 14C10, Cell Signalling Technology, USA), anti–microtubule-associated protein 1 light chain 3B (LC3B; 1:1000, L7543, Sigma-Aldrich, USA), anti-SQSTM1/p62 (1:1000, P0067, Sigma-Aldrich, USA), anti- TGF-beta activated kinase 1 (TAK1) binding protein 3 (TAB3; 1:1000, ab124723, Abcam, UK), or anti-autophagy and beclin 1 regulator 1 (AMBRA1; 1:1000, NBP1-07124, Novus, USA) antibodies at 4°C overnight. Membranes were washed with TBST and incubated with HRP-conjugated secondary antibodies. The enhanced chemiluminescence method (Bio-Rad, Hercules, USA) was used to quantify the Western blot results. All experiments were repeated three times.

Statistical analysis

Experimental data were analysed by GraphPad Prism, and statistical significance was set as a two-sided *P*-value of <0.05. Numerical variables with a normal distribution are presented as the means (S.E.M.)s and were analysed by an unpaired *t* test. Non-normally distributed data are shown as the medians and were assessed with the Wilcoxon rank sum test. Correlations between the expression of plasma miRNAs and clinical indices were analysed by Spearman's Rank Correlation Coefficient (one or two of the variables samples were abnormally distributed) with R package rstatix and correlation.

Results

DE mRNAs in DM plasma EXOs

The EXOs isolated from plasma showed cup-shaped, double-membraned morphological characteristics, as assessed by TEM (Supplementary Fig. S1A, available at *Rheumatology* online); the expected 120 nm diameter, as assessed by dynamic light scattering (Supplementary Fig. S1B, available at *Rheumatology* online); and positive expression of CD63, as assessed by bead-based flow cytometry (Supplementary Fig. S1C, available at *Rheumatology* online).

From the mRNA profile of plasma EXOs, we identified 689 DE mRNAs in DM (with threshold $P<0.05$ and fold change >1.5), including 205 upregulated and 484 downregulated mRNAs (Fig. 1A and B). GO analysis of these DE mRNAs showed that autophagy, process utilizing autophagic mechanism, Wnt signalling pathway and response to IFN-gamma were enriched (Fig. 1C). To infer the kinds of peripheral blood immune cells from which DM plasma EXOs, we applied CIBERSORTx to deconvolute plasma EXO mRNA data. As shown in Fig. 1D, plasma EXOs might be secreted mainly by CD8 T cells, regulatory T cells (Tregs) and resting NK cells (raw data shown in Supplementary Data S1, available at *Rheumatology* online).

DE miRNAs in DM plasma EXOs

In the miRNA profiles of plasma EXOs, we identified 44 upregulated and 9 downregulated miRNAs in DM patients compared with NCs (with threshold $P<0.05$ and fold change >2.0), and 4157 predicted target genes for the upregulated miRNAs and 1257 predicted target genes for the downregulated miRNAs (Fig. 2A and B). GO functional analysis suggested that the following processes were significantly enriched in predicted target genes: autophagy, regulation of apoptotic signalling pathway, Wnt signalling pathway, muscle tissue development and response to TGF-β (Fig. 2C). KEGG analysis indicated that the following pathways were significantly enriched: autophagy, the TGF-β signalling pathway, the Wnt signalling pathway and the FOXO signalling pathway (Fig. 2D (raw data shown in Supplementary Data S2, available at *Rheumatology* online).

Plasma EXOs was one component of plasma, we further matched our 53 DE EXOs-miRNAs with these 43 DE miRNAs in DM plasma (from GSE94535, DM post treatment *vs* DM pre-treatment), and found that three miRNAs (hsa-miR-125a-3p, hsa-miR-1246, and hsa-miR-3614-5p) were significantly changed in both datasets (Fig. 2E). We further validated these three miRNAs in 10 DM patients before treatment, 9 DM patients after treatment and 12 NCs by RT-qPCR. The plasma levels of the three miRNAs were increased in DM patients before treatment compared with NCs, hsa-miR-125a-3p and hsa-miR-3614-5p were significantly decreased in DM patients after treatment; hsa-miR-1246 was also downregulated, but there was no statistically significant difference (Fig. 2F). The predicted target genes of the three miRNAs were enriched in autophagy pathway (Fig. 5A).

Correlations between DM plasma miRNAs (hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p) and clinical indices

We explored the correlations between plasma miRNAs (hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p) and clinical indices. The latter include the demographic characteristics, serological indices, organ involvement and myositis-specific antibodies (MSAs). The level of hsa-miR-125a-3p was positively correlated with the concentrations of aspartate aminotransferase (AST), lactate dehydrogenase (LDH) and alanine aminotransferase (ALT), and negatively correlated with disease duration and the number of neutrophils (NE) (Fig. 3A). The level of hsa-miR-1246 was negatively correlated with the number of monocytes (MO), and significantly positively correlated with the positive rate of the anti-Mi2α (Fig. 3B). The expression of hsa-miR-3614-5p was positively correlated with the concentration of AST, the positive rate of the anti-MDA5 and interstitial lung disease (ILD) (Fig. 3C).

DE lncRNAs in DM plasma EXOs

In the lncRNA profiles of plasma EXOs, we identified 452 DE lncRNAs in DM patients compared with NCs {with threshold $P<0.05$ and absolute value [$\log_2$(fold

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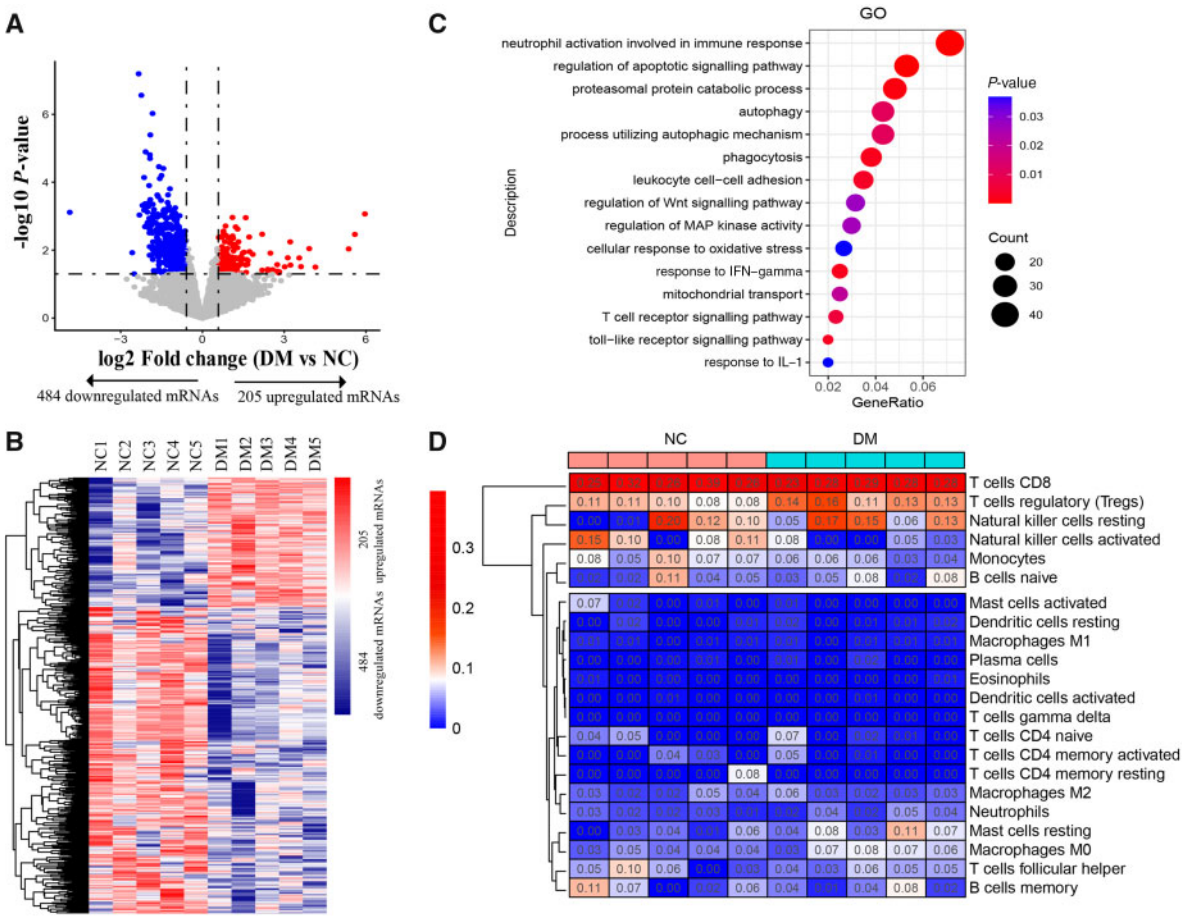
Fig. 1 The DE mRNAs in DM plasma exosomes



A total of 205 upregulated and 484 downregulated mRNAs were identified. (**A**) Volcano plot and (**B**) heatmap. (**C**) GO analysis showed that autophagy and response to IFN-gamma were enriched in DM plasma exosomes. (**D**) CIBERSORTx analysis showed that plasma exosomes were secreted mainly by CD8 T cells, Tregs and NK cells. DE: differentially expressed; GO: gene ontology; NCs: normal controls.

change)] $>1.5\}$. Through bioinformatic analysis, 2267 target genes were predicted for the 313 upregulated lncRNAs, and 897 target genes were predicted for the 139 downregulated lncRNAs (Fig. 4A and B). The GO functional analysis results revealed that the following processes were significantly enriched: autophagy, IFN-β production, IL-12 production and fatty acid transmembrane transport (Fig. 4C). The KEGG analysis results indicated that autophagy, apoptosis, ferroptosis and the JAK-STAT3, FoxO and mTOR signalling pathways were involved (Fig. 4D) (raw data shown in Supplementary Data S3, available at *Rheumatology* online).

To investigate the role of plasma EXO lncRNAs in muscle injury, we further matched our DE EXOs-lncRNAs with the DE lncRNAs in muscle tissue [14]. We found that 9 lncRNAs (ENST00000584157.1, ENST00000523389.1, ENST00000542489.1, ENST00000560054.1, ENST00000428624.1, ENST00000602747.1, ENST00000417650.1, ENST00000608943.1 and ENST00000581240.1) were changed in both datasets, but only three lncRNAs (ENST00000584157.1, ENST00000523380.1 and ENST00000560054.1) were consistently changed (Fig. 4E). The predicted target genes of the three lncRNAs were also enriched in the autophagy pathway (Fig. 5A).

DM Plasma EXOs involved in the autophagy pathway of HSkMCs

From the above results, the predicted target genes of three miRNAs (hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p) and three lncRNAs (ENST00000584157.1, ENST00000523380.1 and ENST00000560054.1) were involved in the autophagy pathway (Fig. 5A), which indicated that these plasma EXO-ncRNAs might participate in the process of DM skeletal muscle damage. Thus, we stimulated HSkMCs with DM plasma EXOs, and found that the expression of hsa-miR-125a-3p, hsa-miR-1246, hsa-miR-3614-5p and ENST00000584157.1 was increased, and the expression of ENST00000523380.1 and ENST00000560054.1 was decreased,

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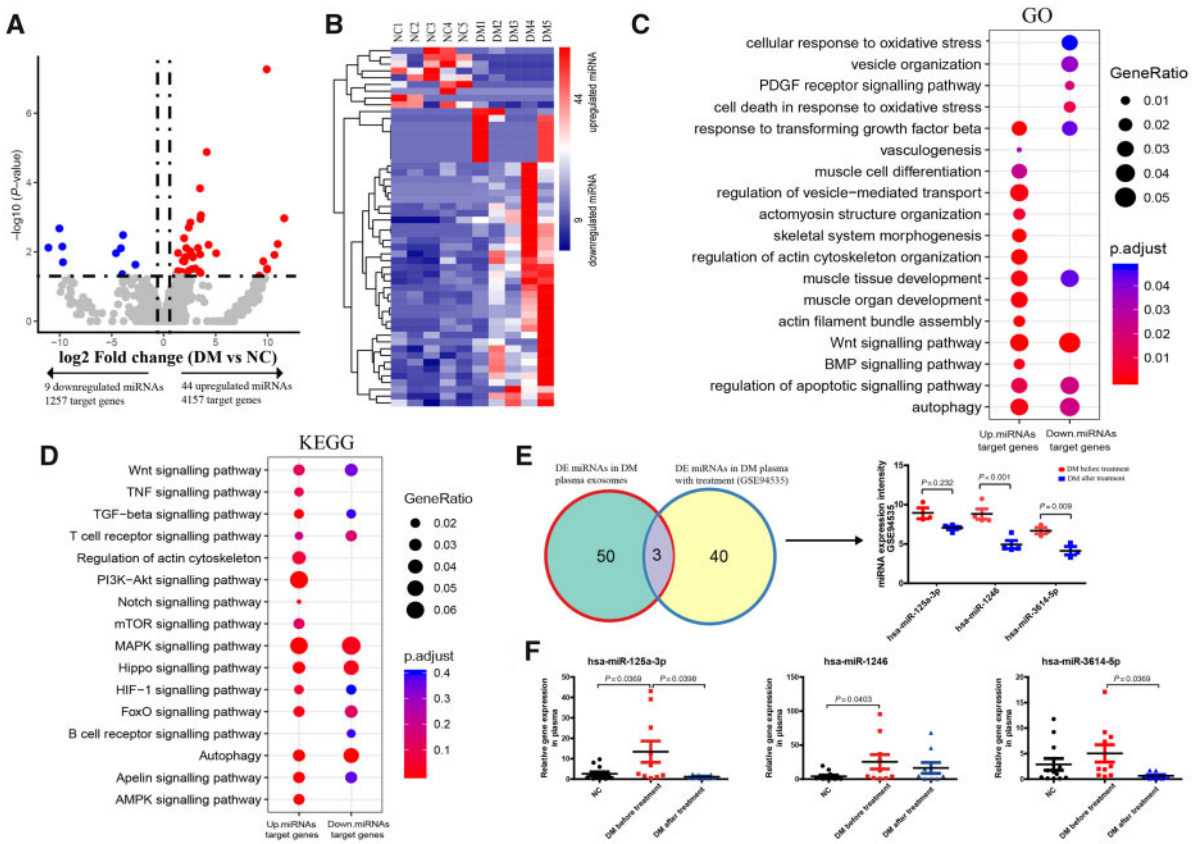
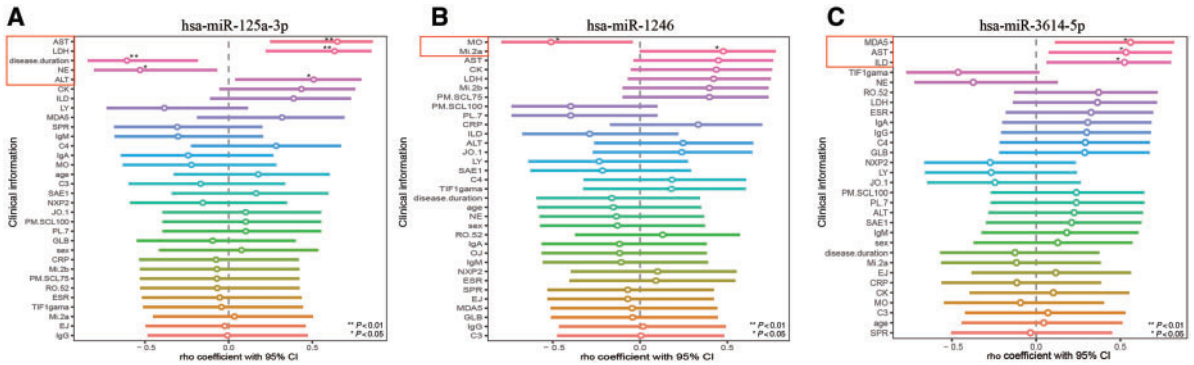
Fig. 2 The DE miRNAs in DM plasma exosomes



Forty-four upregulated miRNAs with 4157 target genes and 9 downregulated miRNAs with 1257 target genes were identified. (**A**) Volcano plot and (**B**) heatmap. (**C**) GO analysis and (**D**) KEGG analysis suggested that the autophagy pathway was significantly enriched in predicted target genes. (**E**) Comparison of our DM plasma exosome miRNA profile with the plasma miRNA profile identified hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p consistently changed in the plasma of DM patients before and after treatment, as shown in the GSE94535 dataset. (**F**) Expression of hsa-miR-125a-3p, hsa-miR-1246 and hsa-miR-3614-5p in the plasma of DM patients and NCs. NCs: normal controls; DE: differentially expressed; GO: Gene Ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes.

Fig. 3 The correlations of DM plasma miRNAs with clinical indices



The plasma expression levels of (**A**) hsa-miR-125a-3p, (**B**) hsa-miR-1246 and (**C**) hsa-miR-3614-5p were correlated with demographic characteristics, serological indices, organ involvement and MSAs. ALT: alanine aminotransferase; AST: aspartate aminotransferase; C3/4: complement 3/4; CK: creatine kinase; GLB: globulin; ILD: interstitial lung disease; LDH: lactate dehydrogenase; LY: number of lymphocytes; MO: number of monocytes; MSAs: myositis-specific antibodies; NE: number of neutrophils.

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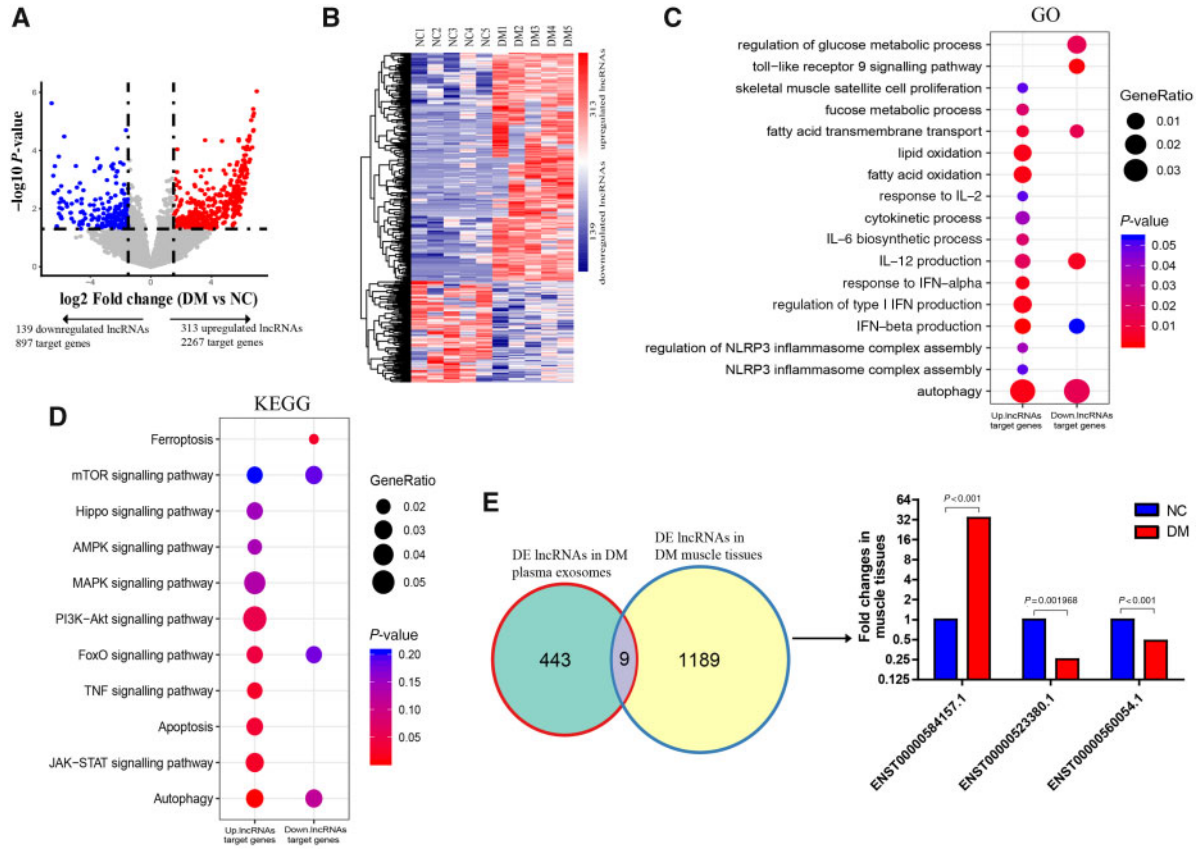
Fig. 4 The DE lncRNAs in DM plasma exosomes



A total of 313 upregulated lncRNAs with 2267 target genes and 139 downregulated lncRNAs with 897 target genes were identified. (**A**) Volcano plot and (**B**) heatmap. (**C**) GO analysis and (**D**) KEGG analysis suggested that the autophagy pathway was significantly enriched in predicted target genes. (**E**) Comparison of the DM plasma exosome lncRNA profile with the muscle tissue lncRNA profile identified three consistently changed lncRNAs [ENST00000584157.1 (increased), ENST00000523380.1 and ENST00000560054.1 (both decreased)]. DE: differentially expressed; GO: Gene Ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes; lncRNAs: long non-coding RNAs; NCs: normal controls.

which was consistent with our RNA sequencing results. The level of the predicted target gene *TAB3* was downregulated and that of the predicted target gene *AMBRA1* was upregulated (Fig. 5B).

Furthermore, we used rapamycin (a classical activator of autophagy) to stimulate HSkMCs. Consistently, the expression of hsa-miR-125a-3p, hsa-miR-1246, hsa-miR-3614-5p, ENST00000584157.1 and *AMBRA1* was increased and that of ENST00000523380.1, ENST00000560054.1 and *TAB3* was decreased (Fig. 5C).

IFN-β is a pro-inflammatory cytokine that reportedly induces muscle impairment and sustains inflammation in DM [17]. Thus, we also stimulated HSkMCs with IFN-β. The levels of the six ncRNAs and two predicted targets genes were similarly changed as above (Fig. 5D).

These results suggested that DM plasma EXOs may participate in the autophagy pathway. We further validated this finding at the protein level. SQSTM1/p62 and LC3B are used as autophagosomal marker proteins. LC3B I is a modified membrane-bound form, the amount of LC3B II correlates with the extent of autophagosome formation, and the LC3B II/I ratio indicates the degree of autophagy. Here, in HSkMCs stimulated with plasma EXOs, LC3B II/I and AMBRA1 protein levels were elevated and SQSTM1/p62 and TAB3 protein levels were reduced (Fig. 5E), indicating that DM plasma EXOs can strongly induce the autophagy process in HSkMCs, playing a role in muscle damage.

Discussion

This study first identified the mRNA, miRNA, and lncRNA profiles of plasma EXOs in adult DM patients. The comprehensive overview of plasma EXOs RNAs gives clues that some ncRNAs may act as biomarkers or be involved in skeletal muscle damage of DM. Moreover, we provided preliminary evidence supporting these results with *in vitro* experiments.

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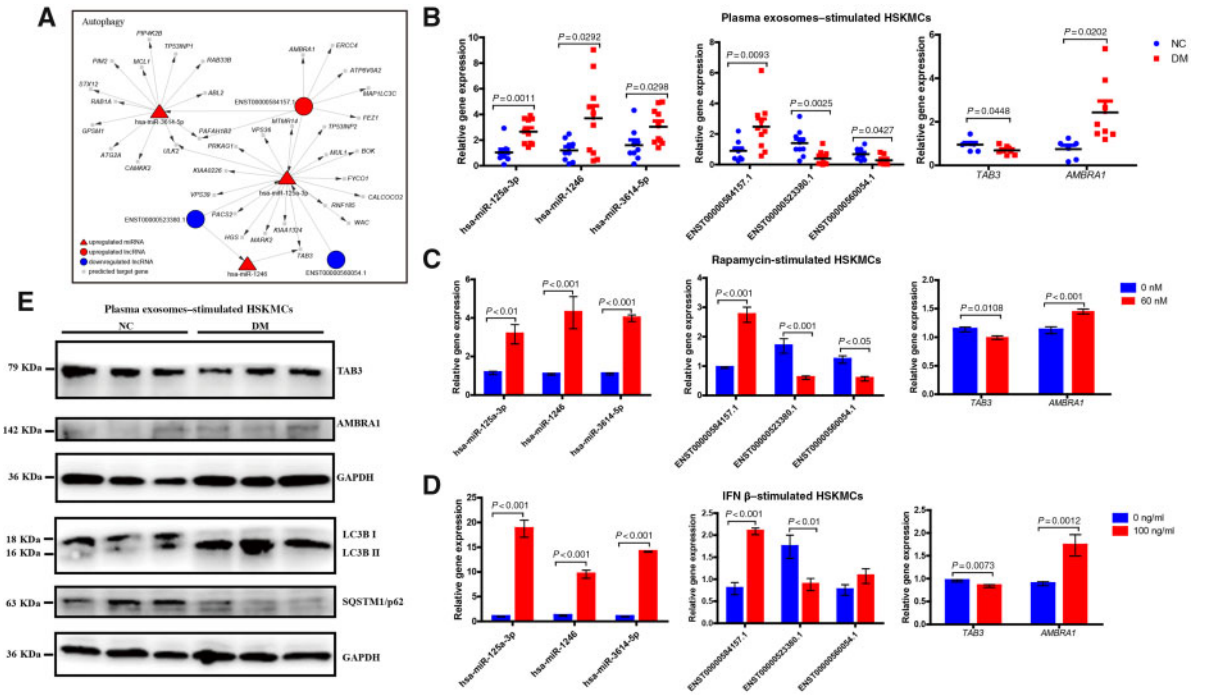
Fig. 5 DM plasma exosomes involved in the autophagy pathway of HSkMCs



(**A**) A lncRNA-miRNA-mRNA interaction network of the autophagy pathway was formed by hsa-miR-125a-3p, hsa-miR-1246, hsa-miR-3614-5p, ENST00000584157.1, ENST00000523380.1, ENST00000560054.1 and their 35 predicted target genes (such as *TAB3* and *AMBRA1*). Validation experiments in HSkMCs stimulated with plasma exosomes from DM patients and NCs showed that (**B**) the expression of hsa-miR-125a-3p, hsa-miR-1246, hsa-miR-3614-5p, ENST00000584157.1 and AMBRA1 was increased and that of ENST00000560054.1 and TAB3 was decreased at the gene level. Similar results were found at the gene level in HSkMCs stimulated with (**C**) rapamycin or (**D**) IFN-β. (**E**) In HSkMCs stimulated with plasma exosomes, the protein levels of LC3B II/I and AMBRA1 were elevated, and those of SQSTM1/p62 and TAB3 were reduced. AMBRA1: autophagy and beclin 1 regulator 1; HSkMCs: human skeletal muscle cells; LC3B: microtubule-associated protein 1 light chain 3B; lncRNAs: long non-coding RNAs; NCs: normal controls; TAB3: TGF-beta activated kinase 1 (TAK1) binding protein 3.

Concerning the DE mRNAs, miRNAs and lncRNAs of plasma EXOs in DM, the enriched pathways and original immune cell-related proportions provided some important information. Accumulating evidence has suggested that type I IFN (IFN1) plays a significant role in regulating both inflammation and immune responses in the development of muscle injury in DM [2, 18]. IFN1 can upregulate the expression of MHC I on muscle fibres and induce the excess production of pro-inflammatory cytokines/chemokines in immune cells [3]. In addition to the common perimysial and perivascular inflammatory infiltrates (such as macrophages, CD20+ B cells, CD4+ T cells, CD25+ plasma cells, and plasmacytoid dendritic cells) [19], CD8+ T cells, Tregs and NK cells may also play important roles, by secrete the majority of plasma EXOs. The decreased function and prevalence of peripheral NK cells and Treg cells have been associated with DM disease activity and treatment effect [20], which may also play a role in the relationship between DM and malignancy [21]. Our results provide insights into the potential functions of circulating CD8+ T cells, Tregs and NK cells in delivering IFN-related mRNAs to recipient cells and in mediating blood vessel and muscle tissue damage by EXOs [18].

According to other studies, hsa-let-7b, hsa-miR-3907 and hsa-miR-4299 in DM serum have been associated with disease activity [22], hsa-miR-4442 and hsa-miR-3187-3p in DM plasma have been associated with drug treatment [23]. We identified and verified hsa-miR-125a-3p, hsa-miR-1246, and hsa-miR-3614-5p in the plasma EXOs and in plasma of DM patients during treatment. These 3 DE miRNAs were correlated with muscle enzymes, organ involvement and MSAs, which may be prognostic biomarkers. Among them, the correlation is stronger between hsa-miR-125a-3p and AST, ALT and LDH than with CK ($P = 0.07$). Histopathologically, DM muscle biopsies are characterized by perifascicular atrophy, fibrosis, CD4+ cells infiltration, and the presence of terminal complement C5b-9 membrane attack complex. Unlike polymyositis, we can't find obvious myofiber damage or necrosis in DM muscle tissues. So we think the levels of CK couldn't totally reflect the disease

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activity of DM. As hsa-miR-125a-3p was increased in pre-treatment DM compared with NC, and decreased in post-treatment DM patients, hsa-miR-125a-3p might associate with the whole body organ damage status in DM patients.

Recent data have suggested that lncRNAs are involved in myopathies and participate in muscle differentiation, regeneration, and function [24–26]. However, few studies have studied lncRNAs and exosomal lncRNAs in patients with DM. In a microarray analysis, ENST00000584157.1 was the most upregulated lncRNA in the muscle tissue of treatment-naïve DM patients compared with that of the NC; linc-DGCR6-1 was highly expressed in the perifascicular areas of the muscle fibres and was implicated in the regulation of the IFN1-inducible gene *USP18* [14]. Here, we identified 3 DE lncRNAs (ENST00000584157.1, ENST00000523380.1, and ENST00000560054.1) in both muscle and plasma EXOs of DM patients. More importantly, the aforementioned 3 miRNAs and 3 lncRNAs formed a network and participated in autophagy.

Autophagy is a cellular process during which autophagosomes form to move unwanted or damaged proteins destined for degradation to lysosomes [27]. Reports have indicated that autophagic processes are active in DM and JDM patients. It shows that the proportions of myofibres containing the autophagy markers LC3B and SQSTM1/p62 and the levels of the potent autophagy inducers TLR4 and TLR3 are higher in DM muscle tissues [28, 29]. On the other hand, IFN-β–induced ROS generation is associated with mitochondrial damage and the activation of autophagy pathways, contributing to muscle impairment and inflammation maintenance in DM muscle tissue [30, 31]. Rapamycin is a macrocyclic triene antibiotic possessing potent immunosuppressant and anticancer activity that inhibits the mammalian target of rapamycin (mTOR), and inhibition of the mTOR pathway by rapamycin blocks the downstream elements in the PI3K-Akt signalling pathway, resulting in autophagy activation. Reports indicate that rapamycin can also successfully induce autophagy in C2C12 myoblast cells, accompanied by a significant increase in the level of the LC3B protein, *in vitro* [16]. From our autophagy network, TAB3 functions as a tonic inhibitor of autophagy [32], and its overexpression can negatively regulate autophagy, resulting in SQSTM1/p62 accumulation independent of TAK1 activity [33]. AMBRA1 is an important upstream regulator of autophagy in the development of different model systems. AMBRA1 binds to beclin 1 and positively regulates the lipid kinase activity of the beclin 1/VPS 34 complex, thus promoting autophagosome formation [34]. In atrophic muscle cells, AMBRA1 can initiate the autophagy response by acting as a cofactor with TRIM32 and ULK1 [35]. In our study, we used plasma EXOs to stimulate HSkMCs and successfully induce autophagy, indicating important mechanisms linking plasma EXOs with DM muscle damage.

This study still has some limitations. First is the small number of plasma EXO samples we studied. We will conduct further studies with a sample size large enough to ensure the homogeneity of NC samples and the heterogeneity of patient samples. This will enable the comparison of EXO data between DM patients from different groups, such as before and after treatment groups and groups with different MSAs.

Conclusion

By identifying the comprehensive RNA profiles of DM plasma EXOs, we verified three miRNAs as potential biomarkers, and some molecules that may be potential targets in the autophagy pathway of muscle damage. These findings provide a basis for more in-depth exploration of plasma EXOs in DM, to improve the understanding of DM pathogenesis and potential applications in disease diagnosis and therapy.

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Data availability statement

The datasets used and/or analysed during the current study are available from the corresponding authors on reasonable request, and are included within the article and its additional files.

Supplementary data

Supplementary data are available at *Rheumatology* online.

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